Appendix to the Manuscript
Enabling Semi-Supervised Travel Mode Prediction Through Synthetic Unlabelled Trip Instances

1 Data used in the experimental part of the study

The survey data were collected by a marketing research company that followed quality control and data protection standards. The research team developed the questionnaires. The contractor was selected through a legally mandated public procurement process. All participants were briefed on the survey's objectives and provided their informed consent for data collection by reading and signing the consent form.

All data sets used in this work rely on the surveys summarised in the main manuscript. Further details on the way the surveys providing data used in this work were performed can be found in [2].

2 Development of semi-synthetic TMC data streams

This section presents a flow diagram, illustrated in Figure 1, that outlines a process for generating semi-synthetic TMC data streams for machine learning applications. This process involves integrating real-world, labelled instances with synthetic data.

The data acquisition phase initiates with the procurement of data from two primary and distinct sources. The first source comprises a dataset of labelled instances, which is subsequently partitioned into two subsets: S_W, representing the comprehensive pool of original labelled instances, and S_T, which specifically designates labelled instances reserved for testing purposes. The second data source is S_G, which constitutes the repository for synthetic data.

The flow diagram delineates the subsequent sequence of operations. The initial phase involves the simultaneous preparation of two distinct data streams: one derived from labelled instances and another from synthetic data. Concurrently, a specific quantity (q) of labeled instances is extracted from S_W and subsequently designated as $S_F(q)$ (Stage 1a). The remaining labeled instances

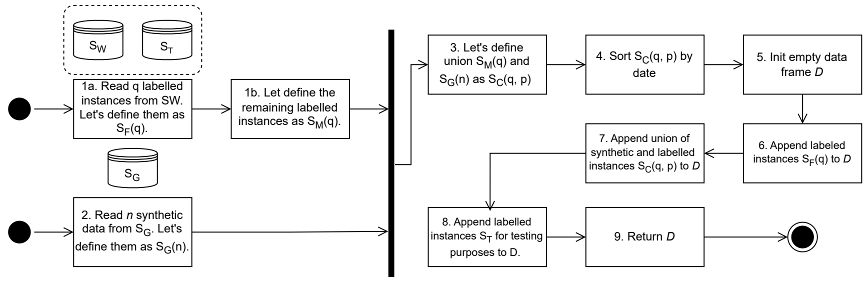


Figure 1: The flow diagram of the development of semi-synthetic TMC data streams.

within S_W, following the extraction of $S_F(q)$, are then defined as $S_M(q)$ (Stage 1b). At the second stage, a quantity n of synthetic data points is retrieved in parallel from S_G and formally defined as $S_G(n)$. Subsequently, at the third stage, the union of the remaining labelled instances, $S_M(q)$, and the synthetic data, $S_G(n)$, is created and designated as $S_C(q,p)$. The fourth stage involves sorting the newly formed combined dataset, $S_C(q,p)$, by date. At the fifth stage, an empty data frame, labelled D_q^p, is initialised. Finally, the construction of the definitive dataset commences. In the sixth stage, the initially extracted labelled instances, denoted as $S_F(q)$, are appended to the data frame D_q^p. Subsequently, at the seventh stage, the sorted union of synthetic and labelled instances, $S_C(q,p)$, is appended to the data frame D_q^p. In the eighth stage, the labelled instances specifically designated for testing purposes, S_T, are appended to the data frame D_q^p.

In essence, the diagram illustrates a process where a portion of real labelled data S_W is initially separated, the remaining real data is then merged with synthetic data S_G, and subsequently, all these distinct data components, alongside a dedicated testing set S_T, are sequentially appended to a primary data frame D_q^p. The explicit sorting step by date suggests that the data may have a time-series or sequential characteristic.

3 Additional summary of the performance of the models

Figures 1, 2, 3, 4 and 5 present in the main manuscript document accuracy changes arising from the use of SU instances compared to the scenario of not using SU instances. As the aforementioned figures concern only some of the methods and settings we provide here figures relying on the same convention, but for all methods and all data sets.

4 Drift detection

Ditzler G. et at. in work [1] elaborates on the various types and characteristics of data drift, with a particular focus on drift induced by changes in probability. Specifically, it distinguishes between real and virtual drift. Real drift involves changes in the posterior probability $p_t(y|x)$ over time, regardless of shifts in the overall occurrence probability $p_t(x)$ within the data stream. Conversely, virtual drift refers to scenarios where the probability $p_t(x)$ of observing a specific instance x varies over time, while the posterior probability of class y $p_t(y|x)$, given instance x, remains constant.

Active learning algorithms are designed for the specific purpose of detecting concept drift, whereas passive algorithms continuously update their models as new data become available, irrespective of whether drift is present. While both active and passive approaches aim to maintain an up-to-date model, they achieve this through distinct mechanisms.

Notably, the variable most frequently affected by drift was duration, with five distinct changes observed in the travel time if performed with public transportation. Conversely, the least frequent detection occurred for the number of transfers, where only a single instance of drift was recorded.

The drift detection analysis was conducted using the CW1 and CW2 datasets. The data in the CW2 dataset originates from a period following the collection of CW1. The results of the drift detection are presented in Tables 6, 7 and 8, with red vertical lines indicating the detected drift points, and a black vertical line marking the end of the CW1 dataset and the beginning of the streaming processing of the CW2 dataset.

References

[1] Ditzler, G., Roveri, M., Alippi, C., Polikar, R.: Learning in nonstationary environments: A survey (11 2015). https://doi.org/10.1109/MCI.2015.2471196

[2] Hassani, A., Nicińska, A., Drabicki, A., Zawojska, E., Sousa Santos, G., Kula, G., Grythe, H., Zawieska, J., Jaczewska, J., Rachubik, J., Archanowicz-Kudelska, K., Zagórska, K., Grzenda, M., Kubecka, M., Luckner, M., Jakubczyk, M., Wolański, M., Castell, N., Gora, P., Skedsmo, P.W., Rożynek, S., Horosiewicz, S.: Air quality and transport behaviour: sensors, field, and survey data from warsaw, poland. Scientific Data **11**(1), 1305 (Nov 2024). https://doi.org/10.1038/s41597-024-04111-4

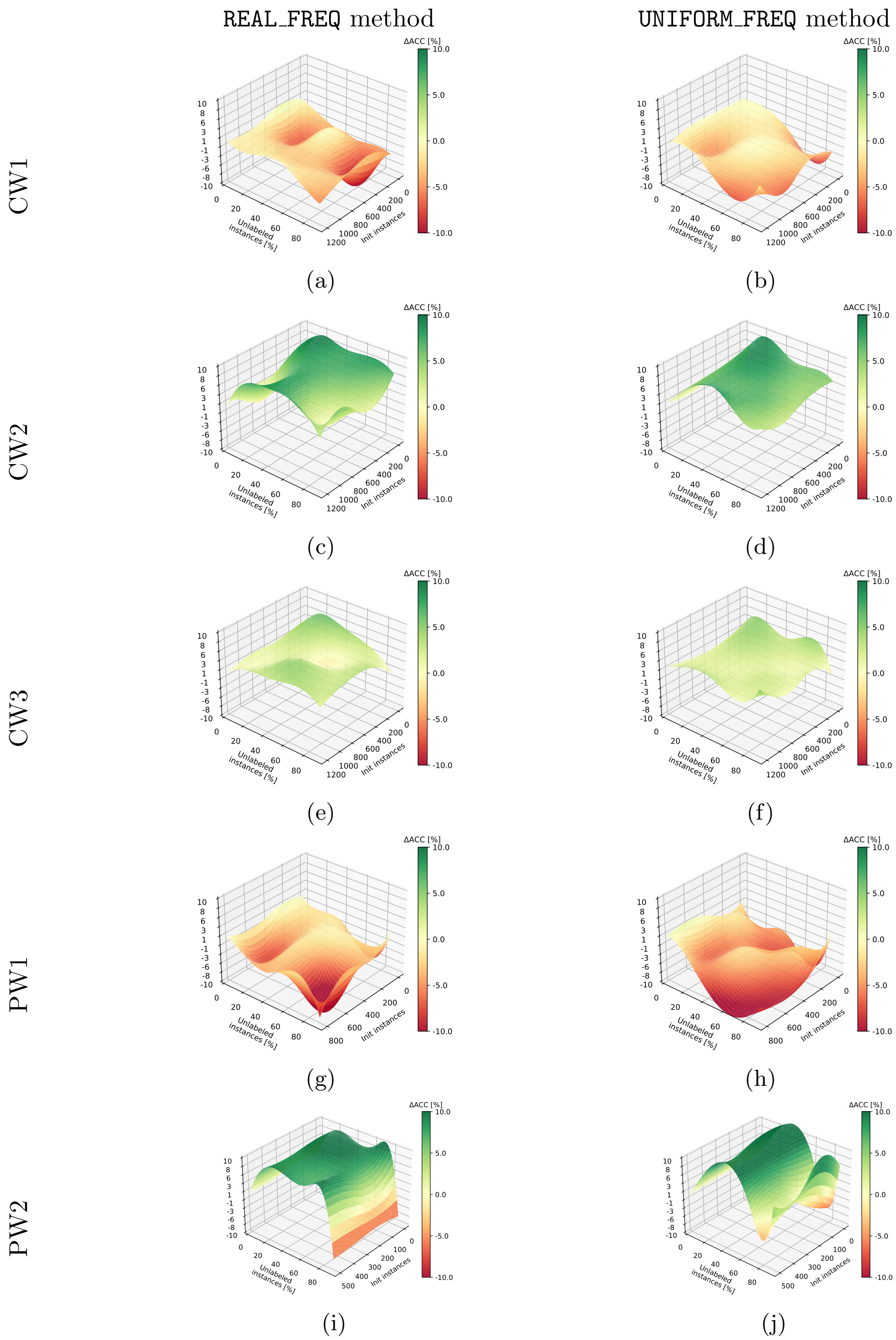


Figure 2: Accuracy change of cluster-and-label (CL) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q. Best viewed in colour.

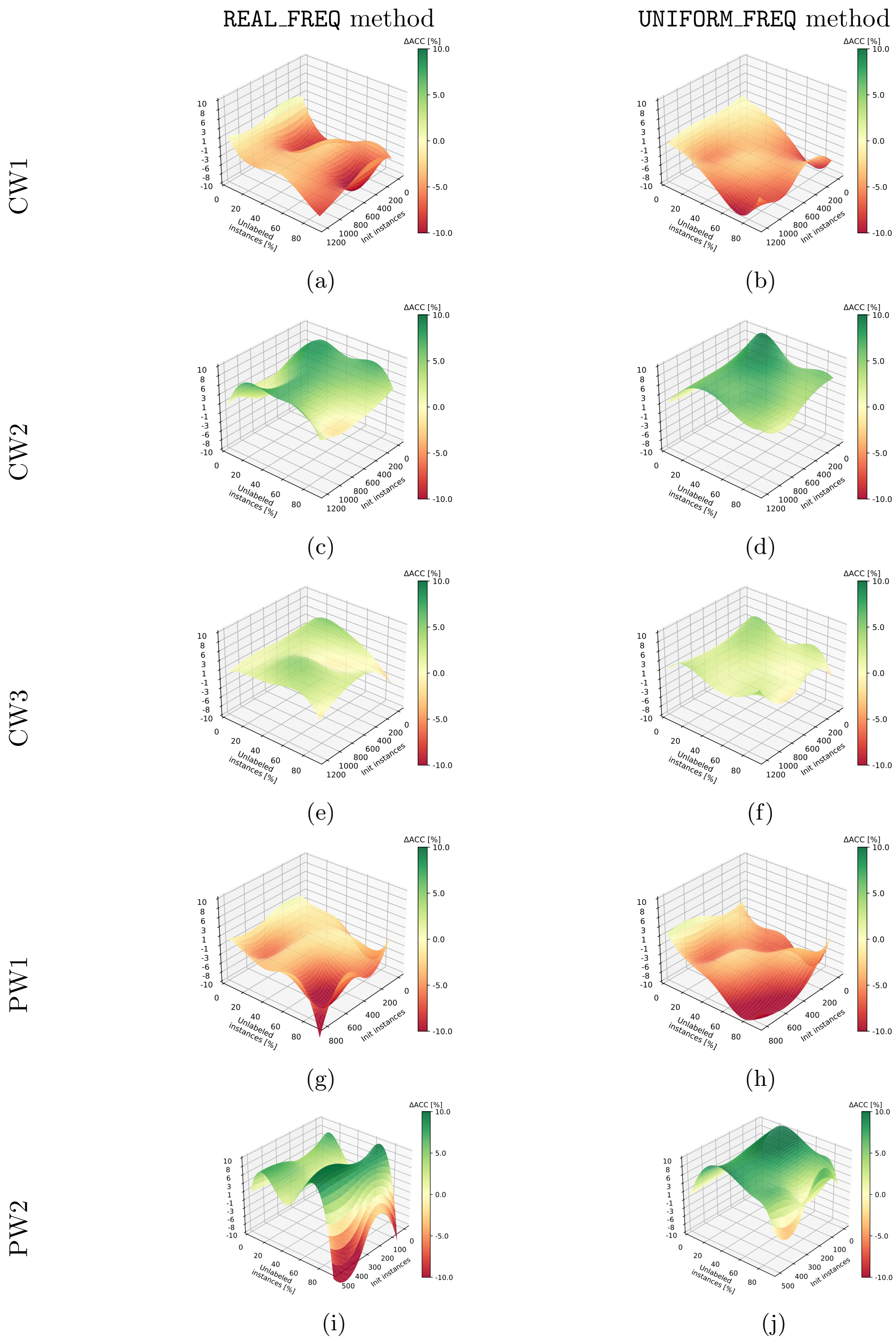


Figure 3: Accuracy change of cluster-and-label with pseudo labelling (CL^L) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q. Best viewed in colour.

REAL_FREQ method | UNIFORM_FREQ method

CW1

CW2

CW3

PW1

PW2

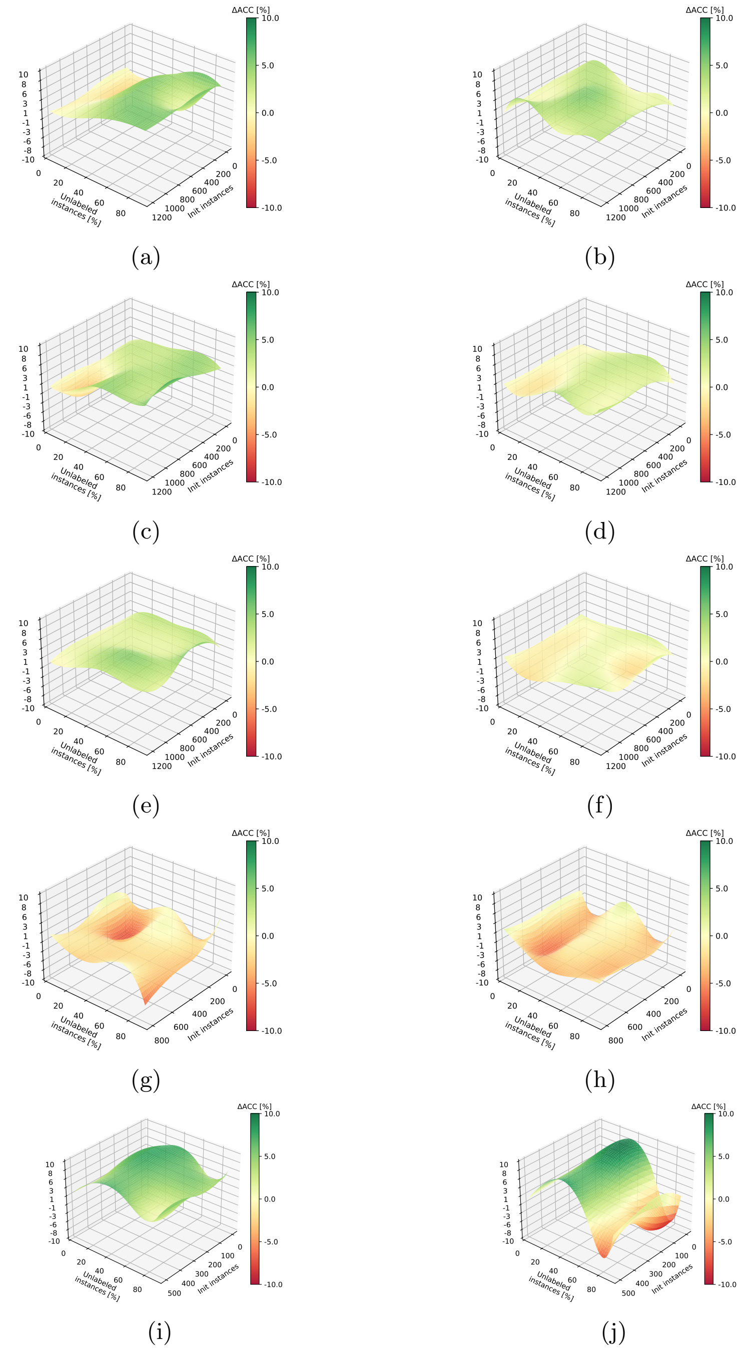


Figure 4: Accuracy change of cluster-and-label with sublearner (CL_{Sub}) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q. Best viewed in colour.

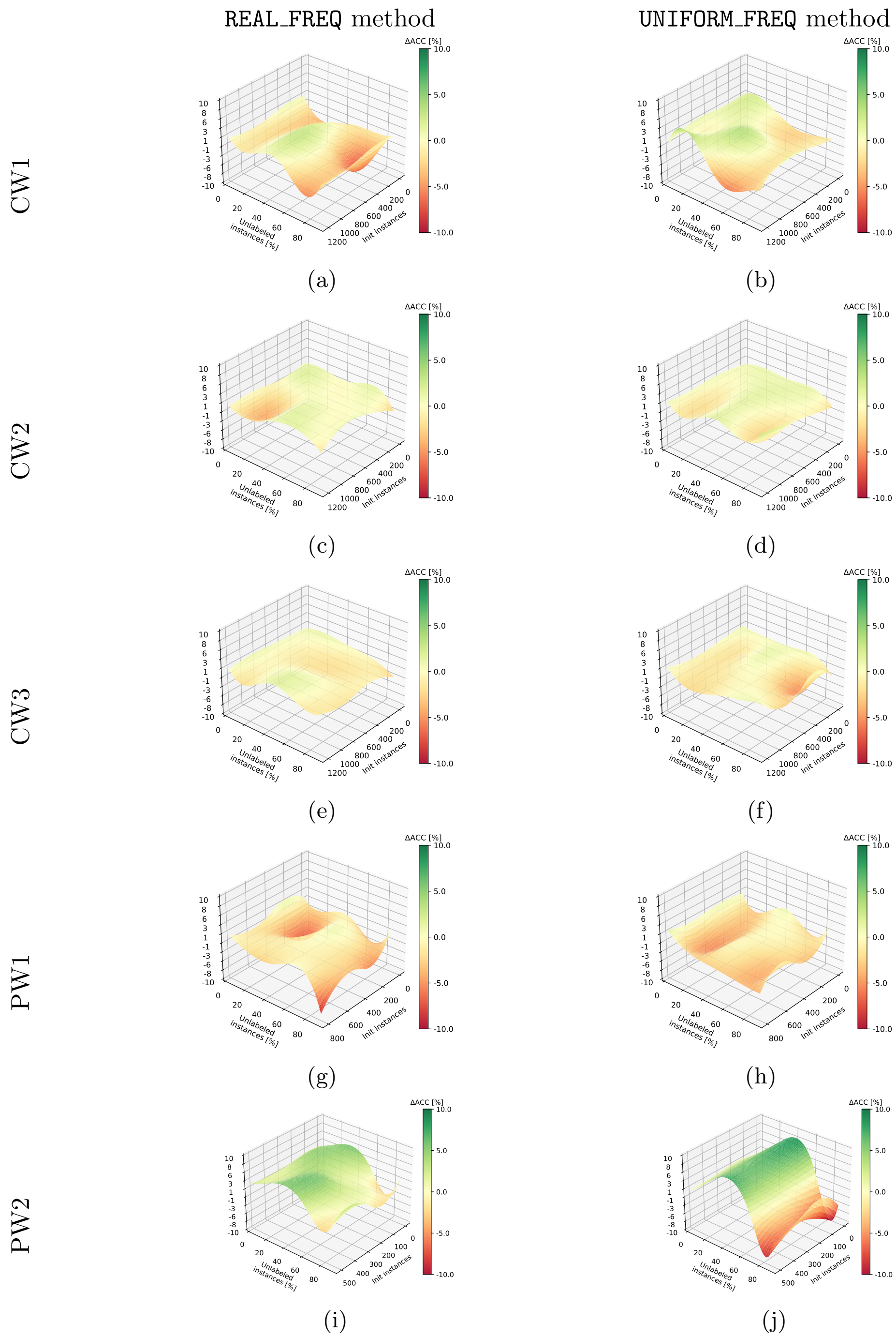


Figure 5: Accuracy change of cluster-and-label with sublearner and pseudo labelling ($\mathrm{CL}^{\mathrm{L}}_{\mathrm{Sub}}$) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q. Best viewed in colour.

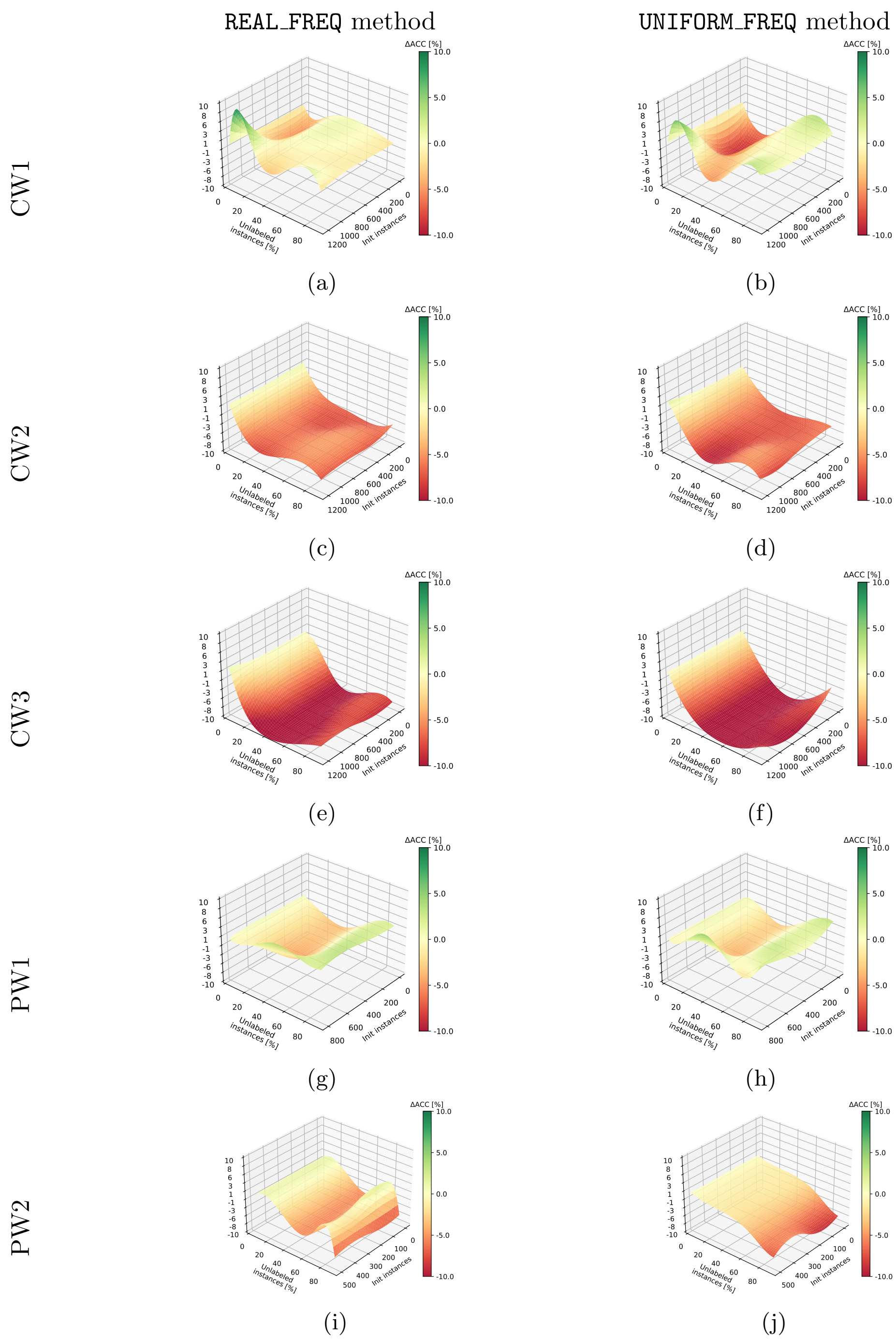
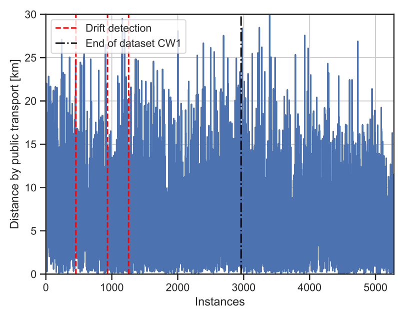
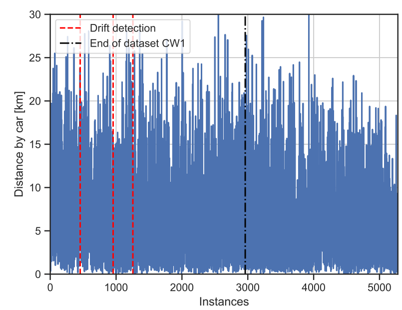


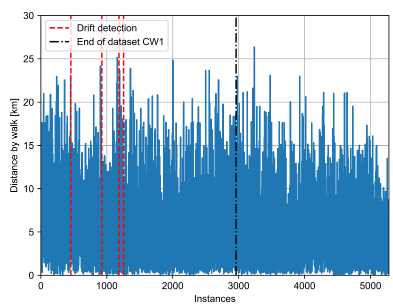
Figure 6: Accuracy change of self-training (SELF) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q. Best viewed in colour.



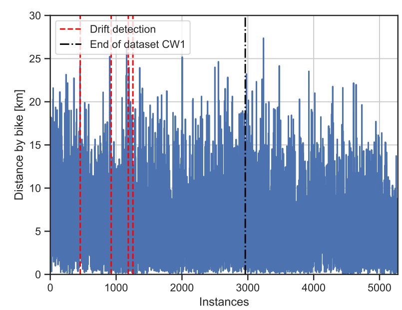
(a) Public Transport Distance



(b) Car Distance

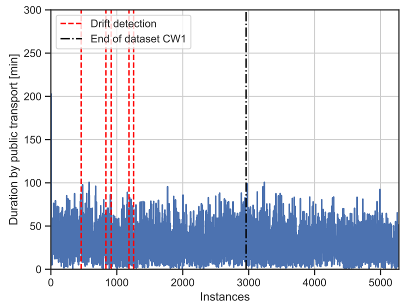


(c) Walk Distance

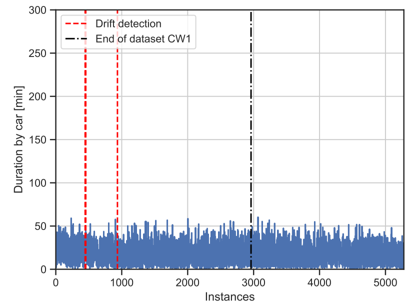


(d) Bike Distance

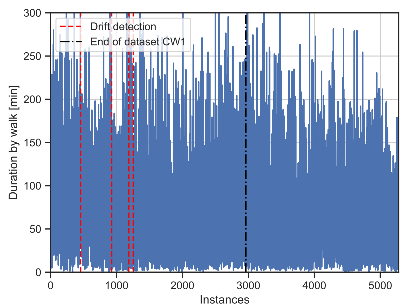
Figure 7: The travel distances recorded in successive surveys over time, with the points of detected drift indicated.



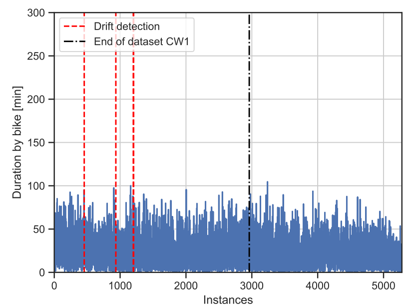
(a) Public Transport Duration



(b) Car Duration

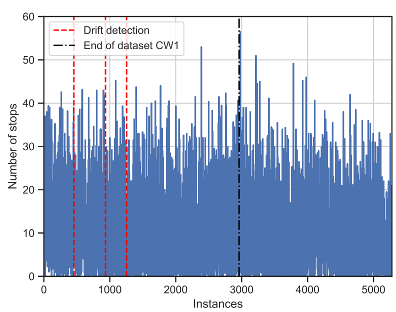


(c) Walk Duration

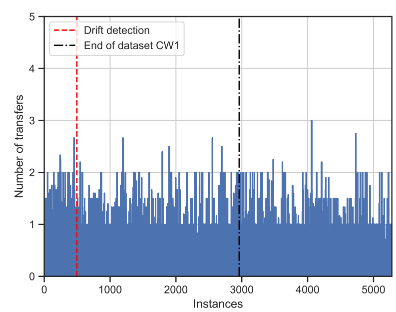


(d) Bike Duration

Figure 8: The travel duration recorded in successive surveys over time, with the points of detected drift indicated.



(a) Number of stops



(b) Transfers number

Figure 9: The number of stops (Fig. 9a) and the recorded instances of transfer (Fig. 9b), as documented in successive surveys over time, with the points of detected drift indicated.